Gonzalo Ortega Carpintero

Curriculum Vitae

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Mathematics PhD Student on Algebraic Topology and Topological Data Analysis

Education

2025 to current **Mathematics PhD student**, *City St George's, University of London*, UK.

- Working on my PhD at Algebraic Topology and Topological Data Analysis supervised by Ximena Fernández.

2024–2025 **Mathematics and Applications master's degree**, *Universidad Autónoma de Madrid*, Spain.

- Studied Analysis, Geometry and Algebra advance courses, while working on my final master's thesis on Topological Data Analysis.

2019–2024 **Mathematics bachelor's degree**, 7.9/10, *Universidad Rey Juan Carlos*, Spain.

- Studied mathematical branches such as Algebraic Topology, Partial Differential Equations, Differential Geometry, Computational Geometry or Complex Analysis.

2019–2024 **Software Engineering bachelor's degree**, 8.25/10, *Universidad Rey Juan Carlos*, Spain.

- Learned several software development skills including multiple programming languages, system design and project management, as well as machine learning fundamentals.

2022 **Software Engineering Erasmus semester**, *Lappeenranta University of Technology*, Finland.

- Semester exchange were I learned about Software Architecture, Computer Networks and Quality Assurance among others.
- Also made university courses about Full-Stack development, Front-End design and Android applications development.

Languages English (advanced C1) and Spanish (native).

Licences Car driving license B2. Boat Skipper.

Experience

Jan 2024 to Apr 2024 **Data scientist intern**, *CSIC, Instituto Cajal, Madrid, Spain.*

- Internship in a neuroscience project, helping to analyze data from multiple mice experiments using machine learning to automatically classify stereotypical behaviors.
- Developing a GUI interface to manage some of the experiment Python controllers.

Jan 2023 to Jun 2024 **Reinforcement teacher**, *IES Blas de Otero, Madrid, Spain.*

- Secondary school teacher in the Comunidad de Madrid program Plan Refuerza, to help teenagers lagging behind in classes to improve their academic results.
- Reinforcement in Mathematics, Physics, Chemistry, Biology, Spanish and English.

Jan 2019 to current **Indie game developer**, *Independent, Madrid, Spain.*

- Self-thought indie game developer, having take part in several online world-wide game jams.
- Projects made using Godot, Unity or Processing with art made in Krita or Aseprite.
- A link to my published games in *itch.io* can be found in the Portfolio section.

Technical skills

Code
- Java, Python, C, Pascal, R, C#, SQL, GDScript, GAMS, LaTeX, Processing.

Tools and libraries
- Visual Studio, PyTorch, Tensorflow, GitHub, Matlab, Google Colab, DeepCutLab, Docker, MySQL, MongoDB, Angular, Spring, Office, Godot, Unity, Android Studio, StarUML, PyCharm, IntelliJ, Krita, Aseprite.

Portfolio

Webpage	gonzaloortega.com
Linkedin	linkedin.com/in/gonzalo-ortega-
GitHub	github.com/Gonzalo-Ortega
itch.io	gonzalo-ortega.itch.io
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